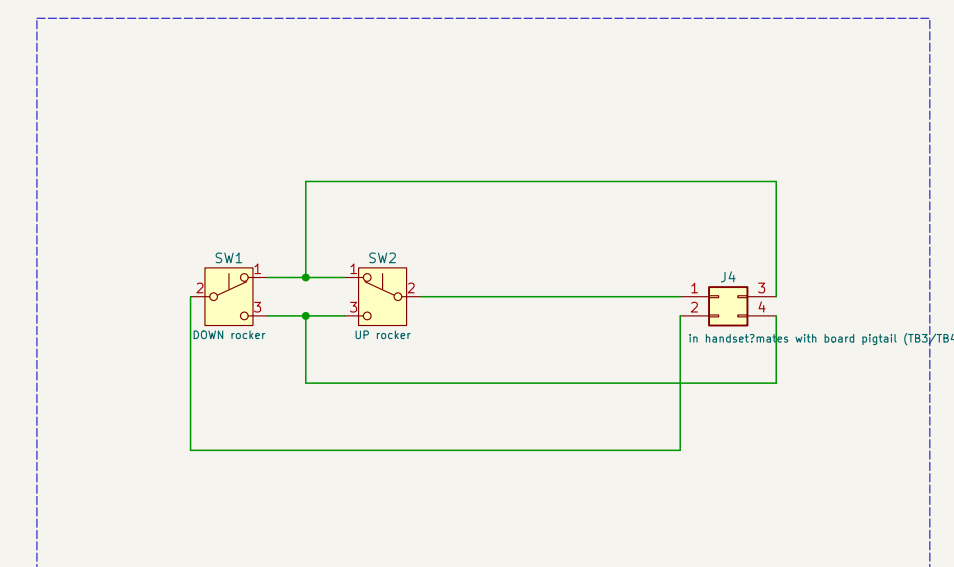
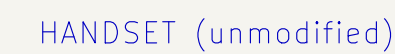
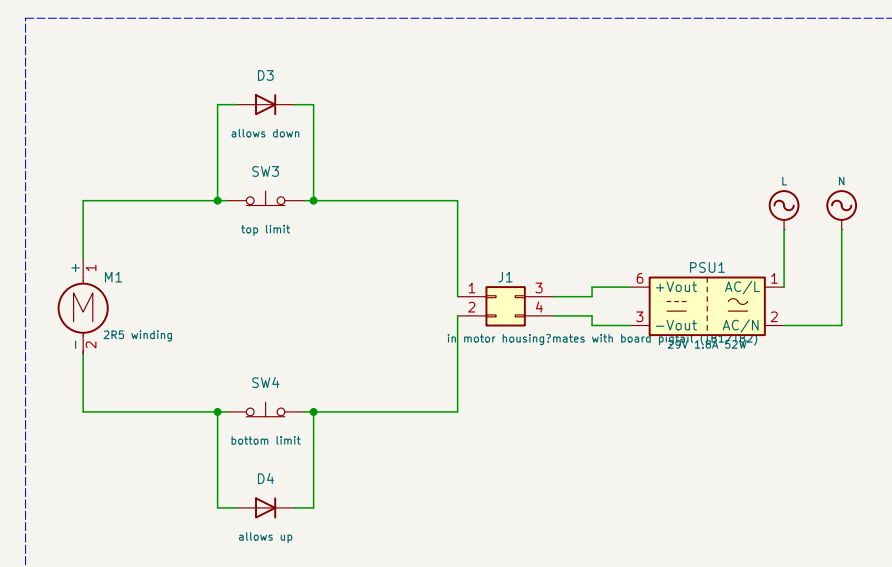
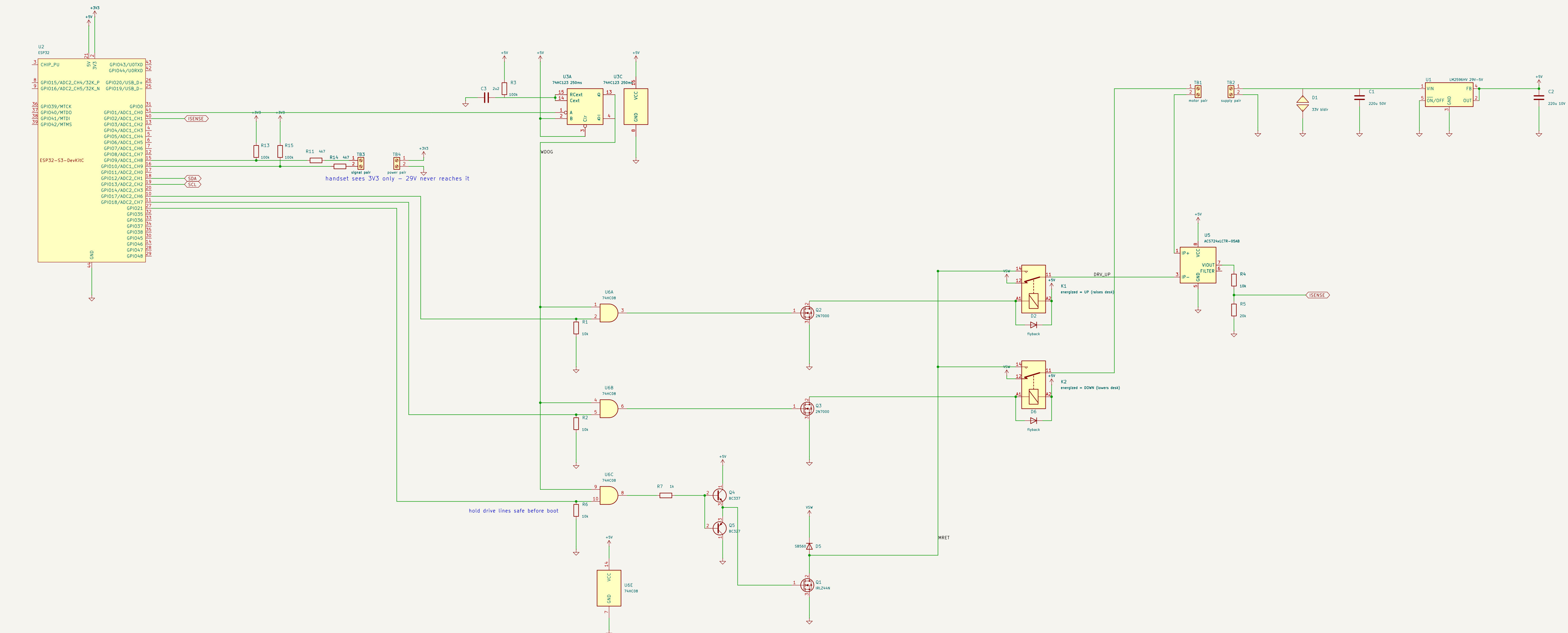




SIMPLE IN-LINE DRIVE. The box sits in the 4-core cable and owns the winding permanently. Reverting is physical: motor – box – handset becomes motor – handset.

The handset is no longer in the motor circuit. It is fed 3V3 and its rockers are read as dry contacts – high at rest through the OEM NC contact, low when pressed. Because 29V can never appear on those conductors, no clamp diodes are needed.

POLARITY (measured on the 4-pin Mini-Fit Jr, circuits numbered on the housing)

circuit 1 = motor circuit 3 = supply +
circuit 2 = motor circuit 4 = supply -
circuit 2 positive, circuit 1 negative = UP. Reverse = DOWN.

K1 / K2 reproduce the OEM rest behaviour:

both off -> both winding ends on VSW = BRAKE
K1 on -> MOT B to return, MOT A stays positive = IIP

K2 on -> MOT_A to return, MOT_B stays positive = DOWN

No combination of relay states can short the supply; there is no shoot-through failure mode.

SEQUENCING: relays change state only with Q1 off and current decayed (~50ms). On a watchdog trip Q1 turns off in microseconds while the relays take milliseconds to release, so the contacts are cold-switched even in a fault. Loss of drive is coast, which is safe because the spindle is self-locking and holds the load unpowered.

US sits in the motor leg so it reads current during freewheel as well as during the PWM on-time, which a low-side shunt would not. ISENSE must be on ADC1 (ADC2 dies with WiFi). R1 / R2 / R6 must sit on non-strapping GPIOs.